

PRODUCT REQUIREMENTS DOCUMENT

Challenges Module

[CUG Challenges](#) · [Individual Challenges](#) · [Rewards & Vouchers](#)

Product	Habit Health — Engagement Module
Feature	Challenges (CUG + Individual) & Rewards System
Document Type	Product Requirements Document (PRD)
Status	FINAL — Ready for Development

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1. Executive Summary

1.1 Problem Statement

Employees need gamified fitness challenges to stay motivated and engaged with their wellness goals. The platform requires a comprehensive challenge system that supports organization-wide competitions (CUG), peer-to-peer challenges (Individual), and a robust rewards system to incentivize participation.

1.2 Solution Overview

- CUG Challenges: Organization-wide step challenges created by admins
- Individual Challenges: Peer-to-peer challenges created by users
- Overall Leaderboard: CUG-wide leaderboard with admin-configurable filters
- Voucher Management: Bulk upload and management of third-party vouchers
- Rewards Configuration: Flexible rule-based reward assignment
- Automated Distribution: 24-hour buffer, email vouchers, wallet credits

2. Scope

2.1 Included in This Release

Feature	Description
CUG Challenges	Admin-created, organization-wide step challenges
Individual Challenges	User-created peer challenges (from Step Sync)
Overall Leaderboard	CUG-wide ranking with admin filter controls
Voucher Upload	Bulk CSV upload of third-party voucher codes
Voucher Management	Track status, expiry, assignment
Rewards Rules	Configurable rules for winner selection
Reward Types	Vouchers (email) and Wallet credits (■)
Auto-distribution	24-hour buffer after challenge ends
Notifications	Push for winning, Email for voucher codes

2.2 Explicitly Out of Scope

Feature	Notes
SPL (Steps Premier League)	Separate PRD — Team-based competition

Internal Vouchers	Only third-party vouchers (Amazon, Flipkart, etc.)
Individual Challenge Rewards	Configurable in future phase
Department Leaderboards	Only individual leaderboards for CUG
CUG Mid-challenge Join	Registration closes when CUG challenge starts
Challenge Cancellation	Admin cannot cancel ongoing challenges

3. Challenge Types Overview

3.1 Comparison

Attribute	CUG Challenge	Individual Challenge
Created By	Admin only	Any user
Scope	All eligible users in CUG	Invited colleagues only
Participation	Opt-in (user joins)	Invite-based
Goal Type	Total steps in duration	Daily goal × Streak days
Duration	Admin-defined (any)	7, 14, 21, or 28 days
Scoring	Most total steps	Streak (tiebreaker: steps)
Tiebreaker	Share rank	Total steps
Rewards	Vouchers / Wallet	Future phase
Leaderboard	Individual ranking	Participants only
Join After Start	No	Yes
Registration End	≤ Start Date	≤ End Date

Note: For CUG Challenges, registration closes when the challenge starts. For Individual Challenges, users can join anytime before the challenge ends (registration_end ≤ end_date).

4. CUG Challenges — User Side

4.1 Challenge Discovery

Users can discover CUG Challenges from the Challenges section on the Engagement homepage, dedicated Challenges listing page, and home page widgets.

Element	Description
Challenge Card	Title, duration, start date, participant count, status badge
Status Badges	UPCOMING / ACTIVE / COMPLETED
Filters	All Joined Upcoming Completed
Search	Search by challenge name

4.2 Join Flow

1. User views challenge details (goal, duration, rewards, rules)
2. User taps "Join Challenge" button
3. System validates: eligibility, registration window, step sync status
4. On success: User added to participants, confirmation shown
5. User can view challenge in "My Challenges" section

4.3 Join Validation Rules

Rule	Behavior
Eligibility	User must be in CUG + User Group
Registration Window	Must join before challenge starts
Step Sync Required	User must have step sync connected
Already Joined	Show "Joined" badge, no duplicate join

4.4 Challenge Detail Page

- Challenge banner and title
- Duration and dates (start → end)
- Goal description (e.g., "Walk the most steps to win!")
- Rewards display (voucher value, wallet amount)
- Current participants count
- User's current rank and total steps (if joined)

- Leaderboard preview (top 5)
- Rules and FAQs tabs

4.5 During Challenge

Feature	Description
Progress Tracking	User sees their total steps and current rank
Leaderboard	Real-time ranking of all participants
Step Sync	Steps synced via Health Connect / Apple Health
Refresh	Leaderboard updates on page open + every 15 minutes
Days Remaining	Countdown to challenge end

4.6 Challenge Completion

When challenge ends, there is a 24-hour buffer period for final step sync. After the buffer, final rankings are calculated and rewards are distributed automatically.

Event	User Experience
Challenge Ends	"Challenge Ended" badge, final sync reminder
Buffer Period (24h)	"Calculating final results..." message
Results Announced	Final rank shown, winner badge for top performers
Reward (Voucher)	Email with voucher code sent
Reward (Wallet)	Amount credited, push notification sent

5. CUG Challenges — Admin Side

5.1 Challenge Creation

Field	Required	Description
Title	Yes	Challenge name (e.g., "March Stepathon")
Description	Yes	Challenge details and motivation
Banner Image	No	Visual for challenge card
Start Date	Yes	When challenge begins
End Date	Yes	When challenge ends
Registration Start	Yes	When users can start joining
Registration End	Yes	Must be $\leq$ Start Date (for CUG)
CUG Assignment	Yes	Select target CUG(s)
User Group Assignment	Yes	Select eligible user groups
Reward Rule	No	Select pre-configured reward rule
Status	Yes	Draft / Published

5.2 Duration Configuration

Admin can set any duration for CUG challenges. There are no fixed duration options. The system calculates duration from start date to end date.

5.3 Edit After Publish

Challenge State	Editable Fields
Before Start	All fields can be edited
During Challenge	Description, Banner, End Date (extend only), Reward Rule
After End	No edits allowed

Note: Admin cannot cancel an ongoing challenge. It must complete naturally.

5.4 View Participants

Admin can view all registered participants with their join date, step sync status, current steps, and rank. Export to CSV is supported.

5.5 Challenge Analytics

Metric	Description
Total Registrations	Users who joined the challenge
Active Participants	Users with steps > 0
Total Steps	Sum of all participant steps
Average Steps	Average steps per participant
Completion Rate	% of users who stayed active throughout
Top Performer	User with most steps

6. Individual Challenges — User Side

Individual Challenges are peer-to-peer competitions created by users to challenge their colleagues. This functionality is accessed from the Activity Challenge page (StepUp Showdown).

6.1 Challenge Creation (User)

Field	Options	Description
Daily Goal	500 - 100,000 steps	Steps per day target
Duration	7 / 14 / 21 / 28 days	Challenge length
Start Date	T+1 to T+7	When challenge begins
Invites	At least 1 required	Colleagues from same CUG

6.2 Scoring System

Individual Challenges use a streak-based scoring system. The winner is determined by the most streak days (days where daily goal was met). If tied, total steps is the tiebreaker.

Metric	Description
Streak Day	Day where user meets or exceeds daily goal
Streak Points	Count of streak days (primary score)
Total Steps	Sum of all steps (tiebreaker)
Winner	Highest streak points; if tied, highest total steps

6.3 Invite Flow

1. Creator searches colleagues by name or email
2. Creator selects one or more colleagues to invite
3. System sends invite notification to selected users
4. Invitees see challenge details before accepting
5. Invitees can Accept or Reject the invite
6. Challenge starts on scheduled date if at least 1 invite accepted

6.4 Join After Start (Individual Challenge)

Unlike CUG Challenges, Individual Challenges allow users to join after the challenge has started. Registration remains open until the challenge end date (`registration_end ≤ end_date`). Late joiners start with 0 steps/streak from their join date.

Scenario	Behavior
Join on Day 1	Full duration, fair competition
Join on Day 5 of 14	Starts with 0, has 9 days remaining
Join on last day	Can still join, limited time to compete
Scoring for late joiner	Streak/steps count from join date only

6.5 Widget States

State	Badge	Display	CTA
Not Connected	Error	"Unable to sync"	Connect Steps
Solo Mode	"Solo Mode"	Steps, Goal, Sync	Take Challenge
Battle Mode Upcoming	"Battle Mode"	Countdown timer	View Status
Invite Received	"New invite"	"[Name] challenged you!"	View Invite
Battle Mode	"Battle Mode"	Steps + mini leaderboard	LeaderBoard

6.6 Challenge Actions

Action	When Available	Effect
Accept Invite	Invite received	Join the challenge
Reject Invite	Invite received	Decline; removed from invite
Modify Challenge	Before start (creator)	Edit goal/duration/date
Quit Challenge	During challenge	Leave; progress lost

7. Individual Challenges — Admin Side

Individual Challenges are user-created. Admin has limited oversight but can configure global settings and view analytics.

7.1 Admin Capabilities

Capability	Description
Enable/Disable	Turn Individual Challenges on/off per CUG
Default Goal	Set default daily step goal per CUG
View All Challenges	See list of all Individual Challenges in CUG
View Analytics	Participation rates, completion rates
Overall Leaderboard Config	Enable/disable and configure CUG-wide leaderboard
Configure Rewards	Future phase: Set up rewards for Individual Challenges

Note: Admin cannot create, edit, or delete user-created Individual Challenges.

8. Overall Leaderboard (CUG-wide)

8.1 Overview

The Individual Challenge page includes an Overall Leaderboard that displays users across the entire CUG. This leaderboard is separate from challenge-specific leaderboards and can be enabled or disabled by admin. Admin can configure filter rules to control which users appear on this leaderboard.

8.2 Admin Configuration

Setting	Options	Description
Enable/Disable	ON / OFF	Show or hide Overall Leaderboard on user side
User Filter	Multiple rules	Which users to display (see below)
Default View	Today / Week / Month / All	Default time filter when page loads
Refresh Rate	Admin-defined	How often leaderboard updates (default: 15 min)

8.3 User Filter Rules

Admin can configure which users appear on the Overall Leaderboard using filter rules. These rules help ensure the leaderboard is meaningful and not cluttered with inactive users.

Filter Rule	Description	Use Case
Steps Synced Only	Only show users with step sync connected	Active participants only
All CUG Users	Show all users in the CUG regardless of sync	Full organization view
Active in Last N Days	Users who synced steps in last N days	Recent activity filter
Minimum Steps Threshold	Only users with $\geq X$ total steps	Filter out zero-step users
User Group Filter	Limit to specific user groups	Department/team leaderboards
Challenge Participants	Only users in active challenges	Engaged users only

8.4 Admin Configuration Interface

1. Admin navigates to Settings → Individual Challenges → Overall Leaderboard
2. Admin toggles Enable/Disable for the leaderboard
3. Admin selects one or more filter rules from dropdown
4. For rules with parameters (e.g., 'Active in Last N Days'), admin enters value
5. Admin sets default time filter view

6. Admin saves configuration

7. Changes apply immediately to user-facing leaderboard

8.5 User-Facing Leaderboard

Element	Description
Toggle Tabs	"All" (CUG-wide) "Yours" (challenge participants)
Time Filters	Today / This Week / This Month / Overall
Ranking	By total steps in selected time period
Display Fields	Rank, Name, Steps, Streak (if applicable)
User Highlight	Current user's row highlighted
Pagination	Load more as user scrolls

Note: When admin disables the Overall Leaderboard, the 'All' tab is hidden from users. Only the 'Yours' tab (challenge participants) remains visible.

9. Voucher Management System

9.1 Overview

Admin can upload third-party voucher codes (Amazon, Flipkart, Swiggy, etc.) via bulk CSV upload. Each voucher code is unique and can only be assigned to one winner (1:1 mapping).

9.2 Voucher Upload Format

Column	Type	Required	Description
voucher_code	String	Yes	Unique voucher code
denomination	Number	Yes	Value in ■ (e.g., 100, 500, 1000)
expiry_date	Date	Yes	Voucher expiration date (YYYY-MM-DD)
provider	String	Yes	Amazon / Flipkart / Swiggy / Custom

9.3 Upload Process

1. Admin navigates to Rewards → Voucher Management
2. Admin clicks "Upload Vouchers" and selects CSV file
3. System validates: format, duplicates, required fields
4. Preview shows valid and invalid rows
5. Admin confirms upload; vouchers added to inventory
6. Success message with count of vouchers added

9.4 Voucher Status Lifecycle

Status	Description
AVAILABLE	Ready to be assigned to winners
RESERVED	Linked to a reward rule (not yet distributed)
ASSIGNED	Sent to a winner
EXPIRED	Past expiry date; auto-removed from available pool

9.5 Voucher Inventory View

- Filter by: Provider, Denomination, Status, Expiry Range
- Search by voucher code
- View total available by denomination

- View assignment history (who received which voucher)
- Export voucher report (CSV)

9.6 Auto-Expiry Handling

System automatically marks vouchers as EXPIRED when they pass their expiry_date. Expired vouchers are removed from the available pool and cannot be assigned to winners. Admin receives a weekly report of expiring vouchers (7-day warning).

10. Rewards Configuration System

10.1 Overview

Rewards are configured as reusable "Reward Rules" that can be attached to challenges. Each rule defines winner selection criteria and reward distribution. Rules have unique codes for identification and can be reused across multiple challenges.

10.2 Reward Rule Components

Component	Description
Rule Code	Unique identifier (e.g., RULE-CUG-001)
Rule Name	Descriptive name (e.g., "Top 3 Winners - ■1000 Voucher")
Challenge Type	CUG / SPL / Individual
Leaderboard Type	Individual Leaderboard / Team Leaderboard
Winner Selection	Rank-based / Top N / Percentage-based
Reward Tiers	Different rewards per rank/tier
Reward Type	Voucher / Wallet / Both

10.3 Winner Selection Methods

Method	Configuration	Example
Single Winner	Rank #1 only	Only the top performer wins
Top N Winners	Specify N (3, 5, 10, etc.)	Top 5 performers win
Percentage-based	Specify % of participants	Top 10% of participants win
Custom Ranks	Specify exact ranks	Ranks 1, 2, 3, 5, 10 win

10.4 Reward Tiers (Rank-based)

Admin can configure different rewards for different ranks. This allows graduated rewards where higher ranks receive better rewards.

Rank	Reward Type	Value
#1	Voucher	■1,000 Amazon
#2	Voucher	■500 Amazon

#3	Voucher	■200 Amazon
#4 - #10	Wallet	■100 credited

10.5 Reward Rule Creation Flow

1. Admin navigates to Rewards → Reward Rules → Create New
2. Admin enters rule name and selects challenge type
3. Admin selects leaderboard type (Individual / Team)
4. Admin configures winner selection method
5. Admin adds reward tiers with amounts
6. System validates: sufficient vouchers available (warning if not)
7. Admin saves rule; system generates unique Rule Code
8. Rule is now available for assignment to challenges

10.6 SPL-Specific Reward Configuration

For SPL (team challenges), admin can configure rewards for both team and individual leaderboards:

Leaderboard	Winner Type	Example Configuration
Team Leaderboard	All players of winning team	■1,000 voucher to each member
Team Leaderboard	Runners-up team	■500 voucher to each member
Individual Leaderboard	Top performer overall	■2,000 voucher
Individual Leaderboard	Top N performers	■1,000 to top 3
Per-Team Individual	Top performer from each team	■500 to best in each team

10.7 Voucher Availability Check

When admin creates a reward rule or assigns it to a challenge, the system checks if sufficient vouchers are available for the expected number of winners. If insufficient, admin is shown a warning but can still proceed (vouchers can be added later).

11. Reward Distribution & Notifications

11.1 Distribution Timeline

Event	Timing	Action
Challenge Ends	T+0	Final sync reminder sent
Buffer Period	T+0 to T+24h	Allow final step syncs
Results Calculation	T+24h	Final rankings computed
Reward Distribution	T+24h	Vouchers/wallet credits processed
Notifications Sent	T+24h	Winners notified

11.2 Voucher Distribution

1. System identifies winners based on reward rule
2. System matches winners to available vouchers (by denomination)
3. Voucher status updated: AVAILABLE → ASSIGNED
4. Email sent to winner with voucher code (fixed template)
5. Voucher assignment logged in system

11.3 Wallet Distribution

1. System identifies winners based on reward rule
2. Wallet credit amount determined from reward tier
3. Amount credited to user's Habit Health wallet
4. Transaction logged with challenge reference
5. Push notification sent to winner

11.4 Notification Types

Notification	Channel	Content
Challenge Joined	Push, In-app	"You joined [Challenge Name]!"
Challenge Starting	Push	"[Challenge] starts tomorrow!"
Challenge Ended	Push	"[Challenge] has ended. Final sync in progress."
Winner Announcement	Push	"Congratulations! You ranked #X in [Challenge]!"

Voucher Reward	Email	Voucher code + redemption instructions
Wallet Reward	Push, Email	"■X credited to your wallet!"

11.5 Voucher Email Template

A fixed email template is used for voucher distribution. Template includes: Congratulations message, Challenge name, Final rank, Voucher provider, Voucher code, Denomination, Expiry date, Redemption instructions, and Support contact.

12. Technical Requirements

12.1 Step Data Source

Challenge step data is sourced from the Step Sync module. Users must have step sync connected (Health Connect on Android, Apple Health on iOS) to participate.

12.2 Data Refresh

Component	Refresh Rate
Step Data	Background sync every 15 minutes
Leaderboard	On page open + every 15 minutes
Participant Count	Real-time on join
Challenge Status	Real-time

12.3 Buffer Period

After challenge end date/time, a 24-hour buffer period allows final step syncs. During this period, the challenge shows "Calculating results..." and leaderboard is frozen except for step updates. After 24 hours, final rankings are locked and rewards distributed.

12.4 Timezone

All challenge dates and times are in IST (Indian Standard Time). Challenge ends at 11:59:59 PM IST on the end date. Buffer period ends at 11:59:59 PM IST the next day.

13. Edge Cases & Error Handling

Edge Case	Expected Behavior
CUG: User tries to join after start	"Registration closed" message
Individual: User joins after start	Allowed; starts with 0 steps from join date
User without step sync tries to join	"Connect step tracking to join" prompt
Two users have exact same steps (CUG)	Both share the same rank
Winners > Available vouchers	Admin notified during rule creation; distribution paused
Voucher expires before distribution	Voucher excluded; admin notified to add more
User leaves CUG mid-challenge	Remains in challenge; can still win
Challenge has 0 participants	Challenge completes with no winners
User's steps don't sync for days	Shown as 0 steps; rank drops accordingly
Admin extends end date	Challenge continues; participants notified
User tries to join twice	"Already joined" message; no duplicate
Overall leaderboard disabled	"All" tab hidden; only "Yours" tab visible

14. Acceptance Criteria

14.1 CUG Challenges — User

- User can view available CUG challenges
- User can join challenge before registration closes
- User cannot join after challenge starts
- User cannot join without step sync connected
- User can view their rank and total steps
- User can view full leaderboard
- User receives notification when challenge ends
- User receives reward notification (voucher email / wallet push)

14.2 CUG Challenges — Admin

- Admin can create CUG challenge with all required fields
- Admin can set any duration (custom dates)
- Admin can assign reward rule to challenge
- Admin can edit challenge before it starts
- Admin can edit limited fields during challenge
- Admin cannot cancel ongoing challenge
- Admin can view participant list with export
- Admin can view challenge analytics

14.3 Individual Challenges

- User can create Individual Challenge
- User can invite colleagues from same CUG
- Invitee can view challenge details before accepting
- Invitee can accept or reject invite
- User can join after challenge starts (before end)
- Creator can modify challenge before start
- User can quit challenge during challenge
- Streak scoring works correctly
- Tiebreaker (total steps) works correctly

14.4 Overall Leaderboard

- Admin can enable/disable Overall Leaderboard
- Admin can configure user filter rules
- Leaderboard shows correct users based on filter
- Toggle between 'All' and 'Yours' works correctly
- 'All' tab hidden when leaderboard disabled
- Time filters (Today/Week/Month) work correctly

14.5 Voucher Management

- Admin can upload vouchers via CSV
- System validates CSV format and duplicates
- Vouchers show correct status (Available/Reserved/Assigned)
- Expired vouchers auto-removed from available pool
- Admin can filter and search voucher inventory
- Admin can export voucher report

14.6 Rewards Configuration

- Admin can create reward rules with unique codes
- Admin can configure winner selection method
- Admin can set different rewards per rank tier
- System warns if insufficient vouchers
- Admin can assign reward rule to challenge

14.7 Reward Distribution

- 24-hour buffer period works correctly
- Vouchers sent via email with correct code
- Wallet credits applied to winner accounts
- Voucher status updated to ASSIGNED
- Winners receive push notification
- Distribution logged for audit

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